

A-level

ENVIRONMENTAL SCIENCE

Questions by topic: Thermal & oil pollution - Set 1

Mark scheme

AQA spec 3.4.3.2.6-3.4.3.2.7; Pollution; textbook pp. 254-262

Marking guidance

- Each part opens with a line in bold saying how its marks are shared out, followed by one sentence on how to mark it.
- The figure in bold at the start of a marking point is the mark that point earns. Lines indented beneath it, starting with a dash, are the detail that belongs to the same mark; a line starting **OR** is another route to it.
- **Also accept** lists other wordings and answers that earn the mark - wording never has to match these notes, only the science. **Do not credit** lists the answers that look right and earn nothing.
- Where a question asks for a fixed number of points (eg *two* reasons), credit only the first points given.
- For levels-of-response questions, determine the level from the descriptors first, then position the mark within the level using the indicative content.
- Calculations: credit correct working (method marks) even if the final answer is wrong; incorrect answers with no working score 0.

Textbook references: every question below cites the printed page numbers in R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018) from which the content is drawn, together with the AQA 7447 specification reference.

Question	Marking guidance	Marks
01.1	1 mark for the point below. 1 As temperature increases the maximum dissolved oxygen concentration decreases (non-linear: falls faster at low temperatures) Also accept • Any other correct, relevant point of the same kind - wording never has to match these notes.	1
01.2	Calculation 1 mark for the correct answer; working does not have to be shown. 1 About 39 °C (accept 38-40) Also accept • The correct answer however it is set out, including a different but valid method. • An answer carried forward from an earlier incorrect step (error carried forward). • An answer rounded to 2 or more significant figures, and the answer written as a fraction, decimal or percentage where the question does not specify. • The value alone - the unit (°C) is already printed on the answer line. Do not credit • A bare wrong answer with no working.	1

Question	Marking guidance	Marks
01.3	Calculation A correct final answer scores full marks on its own. If the answer is wrong, each correct step of working below earns its mark. 1 9.1 - 7.9 (value at 27 °C, accept 7.8-8.0) 1 = 1.2 mg dm⁻³ (accept 1.1-1.3) Also accept • The correct answer however it is set out, including a different but valid method. • An answer carried forward from an earlier incorrect step (error carried forward). • An answer rounded to 2 or more significant figures, and the answer written as a fraction, decimal or percentage where the question does not specify. • The value alone - the unit (mg dm⁻³) is already printed on the answer line. Do not credit • A bare wrong answer with no working.	2
01.4	1 mark each 1 mark for each of the 2 points below. 1 Figure 1 shows the maximum / saturation values – processes that remove oxygen (aerobic decomposition of organic matter / respiration) can give values below the line 1 Higher temperature increases the metabolic rate of organisms and the rate of bacterial decomposition so oxygen is used faster – values above the line are impossible Also accept • Any other correct, relevant point of the same kind - wording never has to match these notes.	2
01.5	Any 3 of 6 1 mark each for any 3 of the 6 points below, in any order, maximum 3. 1 Increased metabolic rate so more energy used / less surplus for growth 1 Eggs develop faster / hatch earlier (out of step with food supply) 1 Non-indigenous warm-water species may thrive and out-compete indigenous species 1 Toxicity of other pollutants is greater as toxins are metabolised faster 1 Reduced resistance to disease 1 Proteins / enzymes denatured if temperature is very high (above about 40 °C) Also accept • Any other correct, relevant point of the same kind - wording never has to match these notes. Do not credit • Points after the first three - the question asks for three, so a list of extras cannot pick up the marks.	3
01.6	1 mark for the point below. 1 Hot water is sprayed from a shower floor in a tall hollow tower – air absorbs the heat and rises, drawing in more air (updraught), so the water returned is cooler (and saturated with oxygen) Also accept • Any other correct, relevant point of the same kind - wording never has to match these notes.	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 256-257; AQA spec 3.4.3.2.6.	

Question	Marking guidance	Marks
02.1	Any 3 of 6 1 mark each for any 3 of the 6 points below, in any order, maximum 3. 1 Discharge of tank-washing water from tankers 1 Oil rig / drilling accidents / leakage during drilling 1 Pipeline leaks 1 Leakage from storage tanks / refinery spills 1 Waste lubricating oil from vehicles / industry washed into drains 1 Other ship accidents (fuel oil) Also accept - Any other correct, relevant point of the same kind - wording never has to match these notes. Do not credit - Points after the first three - the question asks for three, so a list of extras cannot pick up the marks.	3
02.2	1 mark each 1 mark for each of the 3 points below. 1 Oil sticks feathers together so they lose their insulating properties and birds die of hypothermia 1 Feathers lose buoyancy / waterproofing so heavily oiled birds drown 1 Birds swallow toxic oil when preening – time spent cleaning feathers reduces time for feeding themselves / their young Also accept Any other correct, relevant point of the same kind - wording never has to match these notes.	3
02.3	Any 2 of 5 1 mark each for any 2 of the 5 points below, in any order, maximum 2. 1 The surface film reduces the diffusion of oxygen from the atmosphere into the water, so aerobic organisms suffer from low dissolved oxygen 1 Toxic components (benzene, xylene) dissolve and poison plankton near the surface 1 Oil blocks light so algae / phytoplankton cannot photosynthesise 1 Oil masks odours / narcotises so animals cannot find mates or food by smell 1 Shore organisms (molluscs, crustaceans) are covered and asphyxiated Also accept - Any other correct, relevant point of the same kind - wording never has to match these notes. Do not credit - Points after the first two - the question asks for two, so a list of extras cannot pick up the marks.	2

Question	Marking guidance	Marks
02.4	1 mark each 1 mark for each of the 2 points below. 1 Dispersal / dilution by currents and waves is prevented so oil stays concentrated – in open water most spills disperse in days or weeks 1 Inshore areas have more sensitive ecosystems (shores, reefs, mangroves, breeding sites) and more human activities (fishing, tourism) to be affected Also accept - Any other correct, relevant point of the same kind - wording never has to match these notes. Do not credit - quotas described as solving overfishing on their own - undersized fish are still caught and discarded	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 258-259; AQA spec 3.4.3.2.7.	
03.1	1 mark each 1 mark for each of the 5 points below. 1 Double hull: a gap of up to 2 m between the hulls so damage to the outer hull in a collision / grounding does not release oil 1 Inert gas system: cooled engine exhaust gas (no oxygen) fills the space above the oil so an explosive oil vapour / air mixture cannot form 1 Separate ballast tanks: ballast water is not carried in oil tanks so no residual oil is discharged when ballast is pumped out 1 Twin engines / rudders: if one fails the ship can still be steered / propelled and is not carried onto rocks by currents or wind 1 Crude oil washing: sludges are mixed and removed with the cargo so tanks do not need washing with seawater that was then discharged – oily water is unloaded at the terminal Also accept Any other correct, relevant point of the same kind - wording never has to match these notes.	5
03.2	1 mark each 1 mark for each of the 2 points below. 1 Offshore routes keep tankers further from the coast so if there is engine failure there is more time to deal with it before the ship is carried ashore 1 GPS gives accurate navigation – AIS lets ship positions be monitored and coordinated to avoid collisions and groundings Also accept - Any other correct, relevant point of the same kind - wording never has to match these notes. accurate → close to the true value	2

Question	Marking guidance	Marks
03.3	1 mark each 1 mark for each of the 2 points below. 1 The tank stands on an impermeable base surrounded by a wall 1 The volume enclosed by the wall is greater than the tank volume so all the oil is contained if the tank splits / cannot overflow Also accept - Any other correct, relevant point of the same kind - wording never has to match these notes. impermeable → does not let water through	2
03.4	1 mark for the point below. 1 Separates floating oil from drainage water so the oil is retained (for later treatment) and only water flows to the drains / river Also accept Any other correct, relevant point of the same kind - wording never has to match these notes.	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 259-260; AQA spec 3.4.3.2.7.	
04.1	1 mark each 1 mark for each of the 3 points below. 1 Booms are floating tubes with a skirt that restrict the movement of floating oil, keeping it near the source / away from sensitive sites (mangroves, reef, marine park) 1 They work best in sheltered / calm water such as the lagoon 1 Limitation: they do not work in open water where currents or waves are strong – oil can flow under the skirt in a strong current Also accept Any other correct, relevant point of the same kind - wording never has to match these notes.	3
04.2	1 mark each 1 mark for each of the 3 points below. 1 Skimmers: rotating (metal) discs / belts pick up floating oil which is scraped off and stored 1 Absorbent materials (high surface area, eg textile mops, straw, human hair) soak up oil and are then removed 1 Collected oil / absorbents are incinerated or sent to landfill (or oil recycled) Also accept - Any other correct, relevant point of the same kind - wording never has to match these notes. Do not credit 'landfill gas is just carbon dioxide' - it is roughly half methane, a far stronger greenhouse gas	3

Question	Marking guidance	Marks
04.3	1 mark each 1 mark for each of the 2 points below. 1 Dispersants break the oil into an emulsion of droplets, making it more mobile so it spreads through the water column (to the reef / seabed organisms) rather than staying on the surface 1 They do not reduce the amount of oil – the dispersant / oil mixture may be toxic to corals and plankton in shallow, enclosed water Also accept Any other correct, relevant point of the same kind - wording never has to match these notes.	2
04.4	1 mark each 1 mark for each of the 2 points below. 1 Bacteria that break down hydrocarbons digest the residual oil (most rapid where temperature, oxygen and nutrients are high) 1 Accelerated by adding nutrients (nitrates / phosphates) and aerating (ploughing) / irrigating the sediment Also accept Any other correct, relevant point of the same kind - wording never has to match these notes.	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 261-262; AQA spec 3.4.3.2.7.	
05.1	Calculation A correct final answer scores full marks on its own. If the answer is wrong, each correct step of working below earns its mark. 1 $(2900 - 1400) / 1400 \times 100$ 1 = 107% (accept 107.1) Also accept - The correct answer however it is set out, including a different but valid method. - An answer carried forward from an earlier incorrect step (error carried forward). - An answer rounded to 2 or more significant figures, and the answer written as a fraction, decimal or percentage where the question does not specify. - The value alone - the unit (%) is already printed on the answer line. Do not credit - A bare wrong answer with no working.	2
05.2	1 mark each 1 mark for each of the 2 points below. 1 Large decrease from 24 per year (1975-79) to 2 per year (2010-14) 1 Steepest fall between the late 1970s and 1980s – slower decline / levelling off after 1995 Also accept Any other correct, relevant point of the same kind - wording never has to match these notes.	2

Question	Marking guidance	Marks
05.3	Any 3 of 6 1 mark each for any 3 of the 6 points below, in any order, maximum 3. 1 Double-hulled tankers required for all large tankers 1 Improved navigation (GPS, AIS) and offshore shipping routes 1 Inert gas systems reducing explosions 1 Twin engines / rudders reducing groundings after mechanical failure 1 Stricter regulation, inspection and polluter-pays liability (eg after Exxon Valdez) 1 Separate ballast tanks / crude oil washing Also accept - Any other correct, relevant point of the same kind - wording never has to match these notes. Do not credit - Points after the first three - the question asks for three, so a list of extras cannot pick up the marks.	3
05.4	1 mark each 1 mark for each of the 3 points below. 1 Most oil pollution comes from dispersed / diffuse sources not tanker accidents: waste lubricating oil, industrial oil in drains, road runoff 1 Routine operations: small spills, pipeline leaks, storage tank leakage, oil rig accidents (eg Deepwater Horizon) 1 Growth in oil use and transport means more small releases even if large accidents are rarer Also accept - Any other correct, relevant point of the same kind - wording never has to match these notes. diffuse → pollution from a wide area with no single outlet	3
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 258-262; AQA spec 3.4.3.2.7.	
06.1	Correct answer only 1 mark for the correct option. No mark if more than one box is ticked. 1 B - Steam washing (hot water / steam jets kill organisms in the sediment)	1
06.2	1 mark each 1 mark for each of the 2 points below. 1 Chemicals cause the oil molecules to join together / solidify 1 Producing more solid material that can be collected / lifted more easily than liquid oil Also accept Any other correct, relevant point of the same kind - wording never has to match these notes.	2

Question	Marking guidance	Marks
06.3	1 mark each 1 mark for each of the 2 points below. 1 Air contains about 210 cm³ of oxygen per litre whereas water at 20 °C holds a maximum of about 6 cm³ per litre (less than 3% of the amount in air) 1 So a small amount of oxygen-consuming activity (decomposition, respiration) or warming can remove most of the oxygen – many aquatic organisms are near the lower end of their range of tolerance Also accept • Any other correct, relevant point of the same kind - wording never has to match these notes. • deoxygenation → loss of dissolved oxygen from the water • range of tolerance → the range of conditions in which a species can survive	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 254-262; AQA spec 3.4.3.2.6-3.4.3.2.7.	